

# Project 1 Completion Report

## Daily Research Knowledge Graph

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### Goal and current status

Project 1 asks whether a daily agent can collect machine-learning research and organize it as an evidence-backed knowledge graph. The implemented **paper-fetcher-agent** ingests arXiv and enabled public RSS or Atom feeds, extracts grounded features, atomically rebuilds the graph, computes analyses, and persists SQLite plus bounded JSON outputs. GitHub Actions schedules it each day at 13:17 UTC and supports manual runs [\[workflow\]](#).

### Verified August 6 snapshot

| Measure         | Count | Measure                | Count  |
|-----------------|-------|------------------------|--------|
| Paper nodes     | 7,794 | Report nodes           | 29     |
| Blog-post nodes | 906   | Social-post nodes      | 25     |
| Document nodes  | 8,754 | Concept/category nodes | 197    |
| Total nodes     | 8,951 | Typed edges            | 59,256 |

The active snapshot combines 100 Google DeepMind items, 835 Hugging Face items, and 25 public Reddit **r/MachineLearning** posts with the stored arXiv corpus. Every one of the 8,754 document nodes has at least one edge. The database is `data/arxiv_kg.sqlite3`; bounded analysis artifacts are in `output/knowledge_graph/` [\[run summary\]](#).

### How the graph works

Document nodes connect to topics, methods, research goals, datasets, metrics, and arXiv categories. Semantic edges require exact extractor evidence or source metadata; free keywords do not create them. **RELATED\_TO** uses bounded IDF-weighted supported-concept overlap, threshold 0.20, and at most ten neighbors per document. Hubs use weighted PageRank and degree; clusters use related edges scoring at least 0.35. Search and five bounded JSON APIs expose placement explanations, hubs, clusters, and topic trends [\[system report\]](#).

Summaries copy the first abstract or feed-summary sentence and retain provenance, avoiding generated factual claims. Seven-day recent and baseline windows ending August 6 have status **ok**; no topic met both the three-document minimum and positive-growth condition for an emerging-topic label [\[system report\]](#).

### Scope and remaining limits

Public feeds do not cover every blog or social platform; Reddit input comes from one public subreddit. The rules extractor has a finite vocabulary, so a missing edge can mean missing evidence or an unrecognized concept. No human-rated summary study has been run. Trends describe this corpus, not research quality or future impact.

The first August 6 arXiv smoke run found 3,021 submissions beyond the configured 200-paper safety cap. The checkpoint was deliberately rebased to August 6 to keep the corpus bounded, so papers not already stored from July 13 through that rebase time were skipped [\[system report\]](#). Category-classifier predictions are not graph ground truth and do not create graph edges; classification results remain documented in the separate four-page classification report.